

Echinoderms

The starfish and sea urchins we see at the beach belong to an ancient group of sea-dwelling animals known as echinoderms (“ee-KYE-no derms”). Echinoderms have round or star-shaped bodies and feet like tiny suckers, but no heads or brains. Fossils reveal that echinoderms of the distant past were much like those we see today.



Encrinurus fossil



FAMILY FACT FILE

Key features

- Body divided into five equal parts arranged in a circle around a central disk
- Rows of small, suckerlike feet on base
- No front or back and no head or brain

When

Echinoderms first appeared at the start of the Cambrian Period, about 530 million years ago. Over 7,000 species are found in oceans across the world today.

Clypeus

CLY-pee-us

- **When** 176–135 million years ago (Middle to Late Jurassic)
- **Fossil location** Europe, Africa
- **Habitat** Burrows on the seafloor
- **Size** 2–4½ in (5–12 cm) across

Clypeus was a type of sea urchin. Like a modern sea urchin, it had a hard, rounded shell made up of five parts arranged in a star pattern. The shell was covered by spines, but unlike the stiff, pointed spines of many sea urchins, these were soft and hairlike. *Clypeus* found food by burrowing and eating its way through the mud on the seafloor.



Spine bases

Encrinurus

EN-crine-us

- **When** 235–215 million years ago (Middle Triassic)
- **Fossil location** Europe
- **Habitat** Shallow seas
- **Size** Cup 1½–2¼ in (4–6 cm) long

Attached to the seafloor by a stalk, *Encrinurus* used a ring of 10 feathery arms to catch tiny organisms floating past. The organisms, trapped in a sticky fluid, were then swept by tiny hairs toward a central mouth. *Encrinurus* belonged to a class of echinoderms known as crinoids or sea lilies that still exists today.

Pentasteria

PEN-ta-STEER-ee-a

- **When** 203–100 million years ago (Early Jurassic to Early Cretaceous)
- **Fossil location** Europe
- **Habitat** Sand beds
- **Size** Up to 4½ in (12 cm) across

Pentasteria was a starfish that lived during the age of the dinosaurs. It was much like a modern starfish, with five arms, a mouth in the middle of its underside, and two rows of tubelike feet along each arm. Unlike modern starfish, however, it couldn't use its feet as suckers to prize open shells.



Encrinurus trapped food with its sticky arms. The arms could close tightly for protection from predators.

